TED UNIVERSITY

Faculty of Engineering

Department of Computer Engineering

CMPE 491 Senior Project

ARIS

Project Specifications Report

Fall 2025

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1.Introduction

1.1 Description

In modern military operations, effective communication stands as one of the most important elements for mission success. The traditional communication handheld device may reduce the mobility and attention of a soldier in a high-pressure environment. Harsh environmental conditions, background noise, and physical constraints often disrupt the process of communication or make it unsafe to handle the equipment manually. These are reasons why such systems require an intuitive, reliable, hands-free means of commanding and spreading information out in the field.

Smooth communication in dynamic and unpredictable battlefield conditions is paramount for efficiency as well as safety. The soldiers need a system to keep them connected without compromising focus or physical readiness. The motivation for this project lies in addressing these challenges by making use of AI and sensor-based interaction to minimize manual effort. This would increase situation awareness, reduce delays in communication, and facilitate decision-making in difficult environments.

The project aims to design and implement ARIS, an AI-assisted smart helmet communication system. The system will be able to capture multimodal inputs-facial expressions, hand gestures, and predefined voice commands real-time through an embedded camera and microphone. ARIS will translate the captured inputs into command signals using machine learning algorithms, which can be securely transmitted to a central command or nearby units. The development of real-time recognition capability, security in wireless data transmission, and a modular software structure that can be easily expanded with features in the future are the key objectives of this project.

This report provides the specifications and constraints of the proposed ARIS system. Section 1 introduces the concept of the project and deals with its motivation, objectives, and ethical considerations. Section 2 defines the preliminary system requirements, both functional and non-functional. Section 3 provides a list of references that guided both the design and the ethical framework for the project. The report should give a proper overview of what the system will be used for, what not, and in which direction the design should go for the next phases of development.

1.2 Constraints

1.2.1 Economic Constraints

The development of ARIS is limited by both material and financial resources available within a student project. Sensors, processors and communication modules at the military-grade level are costly and hard to access. The prototype should be done with more affordable and available components such as Raspberry Pi. Moreover, open-source software and free AI frameworks will be prioritized to reduce costs associated with licenses.

1.2.2 Environmental Constraints

The ARIS is designed to get both image and voice data from the user for outdoor use which includes various weather conditions. Some environmental factors like lighting and noise may affect the microphone and camera performance. The prototype will not be tested under harsh conditions but should operate in a moderate environment. Additionally, the battery performance may decrease under extreme temperatures or humidity. The system is expected to operate efficiently in moderate outdoor conditions without requiring special thermal protection.

1.2.3 Social Constraints

ARIS provides communication lines to users via AI-driven machine learning. It is critical to know the instructions and predefined commands to use. Also, the system will collect personal visual and audio data of the user. That's why privacy and access control are major social concerns. Proper data handling and encryption protocols have to be ensured. The authorized user who has access to the data will be determined strictly.

1.2.4 Political and Legal Constraints

The ARIS system concept is inspired by military communication needs however real military communication channels, frequencies, and encryption standards are legally restricted and often classified. Because of this the project will operate in a controlled, simulated environment using open and non-classified protocols to demonstrate system functionality. The main purpose is to validate the recognition and data transmission workflow without accessing or interfering with actual military networks.

1.2.5 Ethical Constraints

The system must be developed with a strong commitment to respect, fairness and transparency because it involves processing visual and audio data from individuals. It is important to handle collected data responsibly. The AI models must be trained with multiple datasets to avoid prejudice toward specific facial features or gestures to ensure performance across various users. Moreover, any data used in the project must comply with ethical research standards.

1.2.6 Health and Safety Constraints

The ARIS prototype must ensure the user's physical safety and comfort since it will be mounted on a military helmet and used for long periods. The total additional weight of the system should remain low to avoid neck or head strain. The materials in contact with the skin must be non-irritating and hypoallergenic. The surface temperature of the device should stay within safe limits during continuous operation to prevent burns or discomfort. Since the device will include a microphone and speakers, the feedback sounds must not exceed safe noise levels. Furthermore, the camera and sensors should not block the soldier's vision or movement. All electronic components must be properly insulated to prevent electrical hazards, and the system should shut down safely in case of malfunction.

1.2.7 Manufacturability and Technical Constraints

As a student-developed prototype, ARIS must be built using accessible and affordable components while maintaining technical feasibility. Military-grade hardware is expensive and often restricted, so the system will use commercially available sensors, microcontrollers, and wireless modules. The design should allow modular assembly, enabling parts such as the camera or microphone to be replaced easily. The PCB size, power consumption, and total weight must remain within the limits suitable for helmet integration. All hardware should be compatible with common fabrication methods like 3D printing or standard soldering. Software tools and libraries used during development will be open-source and compatible with standard operating systems to ensure easy integration and future scalability.

1.2.8 Sustainability Constraints

ARIS is expected to function reliably for long periods in the field while minimizing environmental impact and maintenance needs. Rechargeable batteries will be preferred over disposable ones to ensure long-term sustainability. The hardware design will follow a modular structure so that damaged parts can be replaced instead of discarding the entire unit. Materials used for the casing should be durable, recyclable, and compliant with environmental standards. Additionally, software updates and maintenance should be possible without requiring full replacement of the system, supporting the sustainability of both resources and costs over time.

1.3 Professional and Ethical Issues

The development of ARIS introduces several professional and ethical challenges due to its use of visual, audio, and behavioral data from human subjects. Since the system is designed for field communication and involves real-time processing of personal gestures, facial expressions, and voice commands, respecting user privacy and ensuring data security are fundamental ethical priorities. Any data collected during testing must be handled transparently and stored securely, following strict access control policies. Only authorized team members should have permission to view or process the raw data, and all datasets must be anonymized before being used for training AI models.

Another important ethical concern is bias in AI recognition. Because ARIS relies on machine learning for gesture and facial interpretation, it must be trained with diverse datasets to prevent unfair behavior toward users with different facial features, skin tones, or physical conditions. Ensuring this fairness will help maintain reliability and trust in real-world applications. The team must also remain transparent about the system's accuracy limits and avoid presenting ARIS as a fully autonomous or error-free device.

From a professional standpoint, the team must follow responsible engineering principles throughout the project. All design choices, including component selection and algorithm implementation, should consider safety, maintainability, and long-term usability. Clear documentation must be kept to ensure that future developers can understand, verify, and reproduce the system's processes. Ethical development also requires open communication about the system's purpose: ARIS is a research-oriented prototype, not a military-grade communication tool, and it will operate only in a simulated environment with open communication standards to prevent any legal or security risks.

The project also raises concerns about informed consent. Participants or testers must be aware that their image or voice may be recorded for research purposes, and written consent should be obtained prior to data collection. No personal data should be shared outside the project scope, and all recordings must be deleted after analysis. The system should not perform continuous surveillance or monitoring without user awareness.

Lastly, as engineers, the team must balance innovation with ethical responsibility. The main goal of ARIS is to explore how multimodal AI systems can assist communication safely and efficiently. Every design decision should reflect respect for user privacy, data protection, and human safety. The team should ensure that the project's outcomes serve technological progress while maintaining professional integrity and ethical awareness at every stage of development.

2. Requirements

2.1 Functional Requirements

2.1.1 Input Processing

- The system shall continuously capture visual input from the camera to detect facial expressions and hand gestures
- The system shall capture the audio input from the user through the built-in microphone for voice command recognition.
- The system shall perform real-time detection of visual and audio inputs with minimal latency to ensure uninterrupted operation.
- The system shall transmit the input data to the AI module for processing.

2.1.2 Command Recognition

- The AI module shall match the visual input with a predefined set of command templates stored in the system database.
- The AI module shall compare the audio input with predefined voice command patterns using speech recognition algorithms.
- The system shall assign a corresponding command to each recognized gesture, facial expression or voice input.
- The system shall send the validated command output to the communication module in a structured data format.

2.1.3 Communication Module

- The system shall support both receiving and sending modes to allow two-way communication between the control center and other units.
- The system shall transmit validated commands to the central unit or target device through a secure wireless communication channel.
- The system may later be extended to support direct device-to-device communication if the design allows it.
- The system shall use encryption protocols to ensure data transmission is confidential and secure.

2.1.4 User Interface

- The system shall provide a shared web interface allowing all connected helmets or devices to communicate via a central server.
- The system shall include a master (root) user role responsible for monitoring connections, managing permissions, and ensuring fast, live response.
- The user interface shall present system status and connection time in a simple and readable layout.
- The system shall support adaptability to different environments (military or civilian) without reconfiguration.

2.2 Non-Functional Requirements

2.2.1 Performance

- The system shall process inputs within 10 seconds.
- The system shall achieve a minimum recognition accuracy of 90% under normal lighting and noise conditions.
- The system shall operate continuously for at least 1 hour without overheating or performance degradation.
- The system shall detect and analyze input and errors in real time, minimizing delays between user action and feedback.
- The system shall support multiple concurrent users connected to the same central server without loss of stability.
- ARIS shall maintain stable operation during communication, minimizing crashes or unresponsive behavior.

2.2.2 Security and Privacy

- The system shall ensure mutual authentication between sender and receiver units before communication begins.
- All transmitted data shall be encrypted using secure algorithms (e.g., AES-256).
- No personal or usage data shall be stored or transmitted without explicit user authorization.
- The server environment shall prevent unauthorized access or data leakage through secure configuration and encryption.

2.2.3 Usability

- The system shall operate while the user wears protective gear such as gloves or helmets.
- The user interface shall clearly represent system status through intuitive indicators (e.g., color codes, icons).
- Basic user training or adaptation time shall not exceed 10 minutes for standard command usage.
- The interface shall be designed for simplicity and accessibility for non-technical users.

2.2.4 Reliability

- The system shall detect and recover from minor communication errors automatically.
- The system shall have an uptime of at least 90% during the operation process.
- The connection mechanism shall automatically retry or re-establish links when a failure occurs.

2.2.5 Portability

- The system shall require no modification to its source code when transferred between supported platforms.
- The software should be deployable on multiple hardware platforms, including Raspberry Pi.
- The AI models and configuration files shall be easily updatable without reconfiguration.
- The system shall be adaptable to both military and civilian environments without structural changes.

2.2.6 Maintainability & Administration

- The system shall be easy to deploy, maintain, and update minimizing technical complexity for new setups.
- The deployment process shall require minimal configuration so that it can be installed without expert support.
- The system shall provide basic monitoring capabilities for the root user.
- The AI model and the command dictionary shall be easily updatable without altering the core system architecture.

- The system shall track error occurrences and store their frequency to support maintenance planning.

3. References

1. Association for Computing Machinery. (2018). *ACM code of ethics and professional conduct*. ACM. https://www.acm.org/code-of-ethics
2. Institute of Electrical and Electronics Engineers. (2020). *IEEE code of ethics*. IEEE. https://www.ieee.org/about/corporate/governance/ethics.html